

CAPSTONE · SUPPLY CHAIN ANALYTICS

Late Delivery Risk Prediction

Can we predict which DataCo orders will arrive late – and what should we do about it?

65,752
orders analysed

54.8%
delivered late

0.77
model ROC-AUC

01 · 2 MIN
Purpose & objectives

02 · 2 MIN
Data & exploration

03 · 6 MIN
Five objectives

04 · 3 MIN
Predictive model

05 · 2 MIN
Findings → Q&A

Late Delivery Risk – Dashboard

65,752 orders · Jan 2015 – Jan 2018 · slicers shown at "All" – they filter every order-based visual in the Power BI version

Late Delivery
RISK PREDICTION

Year: 2015 – 2018

Market: Africa | Europe | LATAM | Pacific Asia | USCA

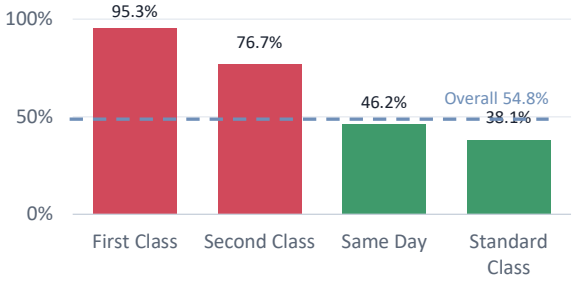
Shipping Mode: First Class | Second Class | Same Day | Standard

Customer Segment: Consumer | Corporate | Home Office

Orders	Late %	Latest Month Late %	Avg Scheduled Days	Avg Actual Days
65,752	54.8%	56.3%	2.93	3.50

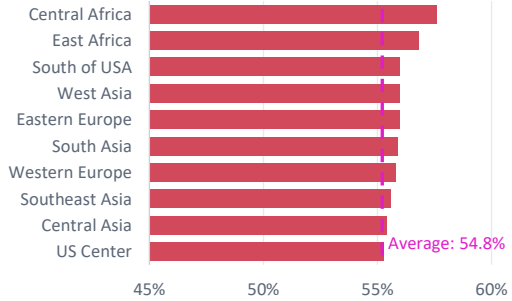
1 · Late % by shipping mode

Late % by Shipping Mode



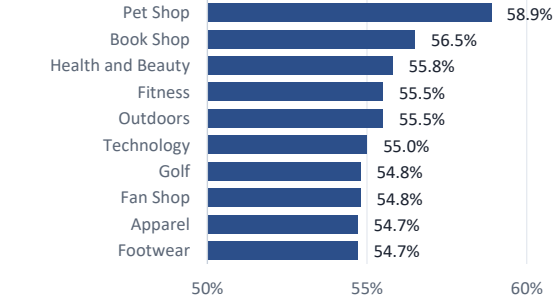
2 · Late % by region

Late % by Order Region (top 10 of 23)



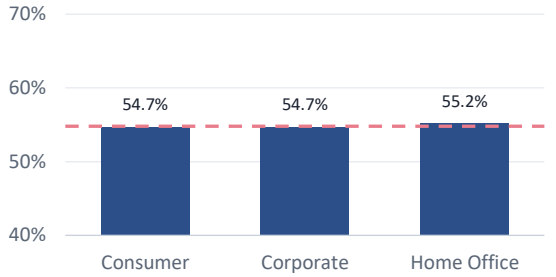
3 · Late % by department

Late Items % by Department Name



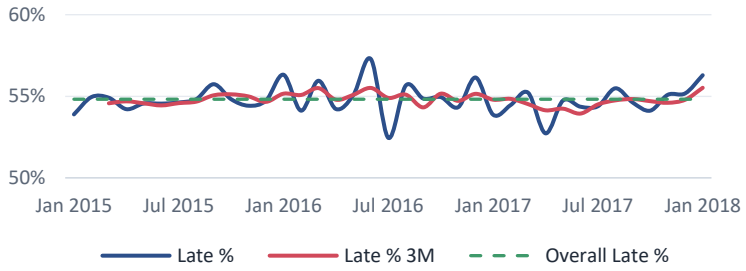
4 · Late % by customer segment

Late % by Customer Segment



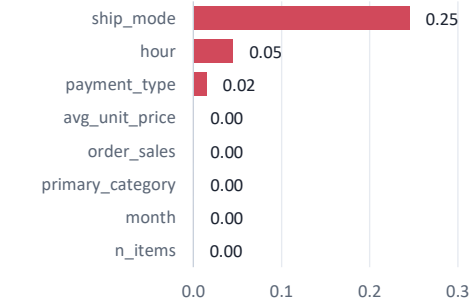
5 · Monthly late % vs average

Late %, Late % 3M and Overall Late % by Month Start



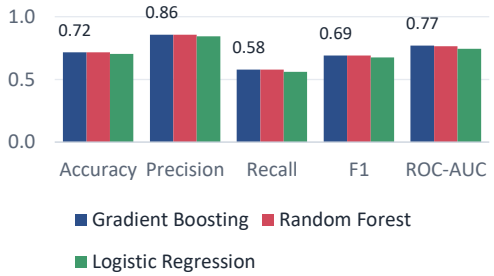
Model drivers – feature importance

Sum of importance by feature



Model comparison (test set)

Sum of Score by Metric and Model



Confusion matrix – Gradient Boosting

Test set · 13,151 orders

Actual Label	Pred. late	Pred. on time	Total
Late	4,170	3,040	**7,210**
On time	698	5,243	**5,941**
Total	**4,868**	**8,283**	**13,151**

Takeaway: shipping mode is the only strong driver (First Class 95.3% late vs Standard 38.1%). Fix the promise on fast modes and score orders at checkout; region, category, segment and timing fixes won't pay off.

Purpose and problem statement

Late delivery hurts customer satisfaction, cost and repeat business

Purpose

1 **Understand**
Where and why late deliveries happen in DataCo's supply chain

2 **Predict**
Build an early-warning model that flags risky orders at order time

3 **Act**
Give operations clear, evidence-based actions

Problem statement

*"How can we analyse and predict whether an order is at risk of late delivery based on **shipping mode**, **geography** and **product category**?"*

SCALE OF THE PROBLEM

54.8% of all orders arrive late

TAKEAWAY **54.8%** of all orders arrive late – a systemic issue, not an edge case.

Research objectives

Late Delivery
RISK PREDICTION

Five descriptive questions, then a predictive model

OBJECTIVE 1 01

Shipping mode

How do shipping modes influence late delivery?

OBJECTIVE 2 02

Geographic risk

How does risk vary across regions, countries and cities?

OBJECTIVE 3 03

Product category

Are some categories or departments more often late?

OBJECTIVE 4 04

Customer segment

Does risk differ by segment or customer value?

OBJECTIVE 5 05

Temporal trends

Is risk cyclical, improving or worsening?

PREDICTION →

Late-order model

Which orders are at risk at order time, and why?

TAKEAWAY Objectives 1–5 are each tested with χ^2 and Cramér's V; the model then checks whether the same drivers matter for prediction.

Dataset and preparation

DataCo supply chain tables merged into one order-level table

65,752
orders analysed

180,519
order items

2015–2018
Jan 2015 – Jan 2018

54.8%
late rate

Sources

- Orders, Shipping, Geography, Payment Types, Customers
- Order Items → Products → Categories → Departments

Raw tables → **Clean & merge** → **Features** → **Model**

Preparation

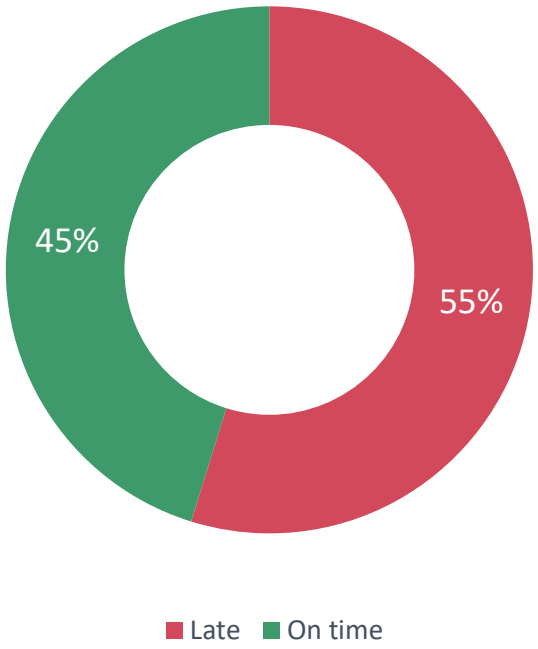
- Customer personal data removed
- Mixed date formats parsed; delay = actual − scheduled days
- Order-level features and customer value tier built
- Only Order Zipcode had material missing values

TAKEAWAY The late flag matches “actual > scheduled days” in **97.5%** of orders – a genuine delivery outcome.

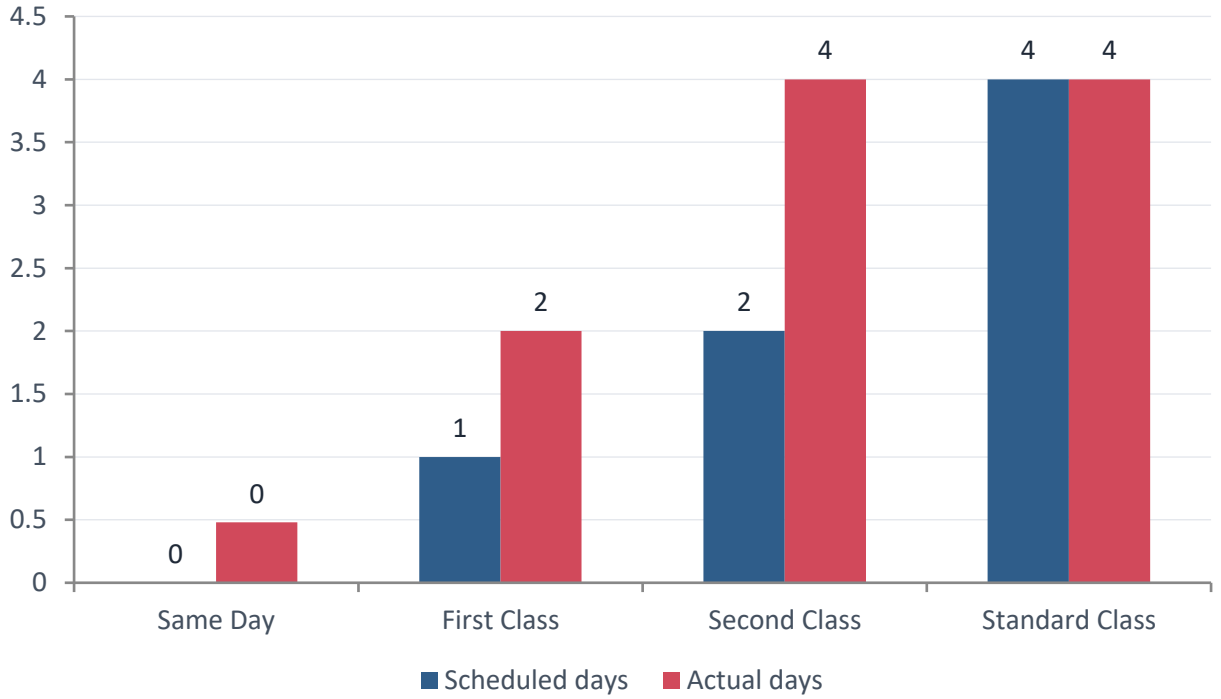
Exploratory analysis

The target is balanced, and faster modes run later than promised

Target balance (% of orders)



Average days by shipping mode



TAKEAWAY About 55% late versus 45% on time – a balanced target. First and Second Class take 1–2 days longer than scheduled.

Objective 1: Shipping mode drives lateness

% of orders delivered late by shipping mode



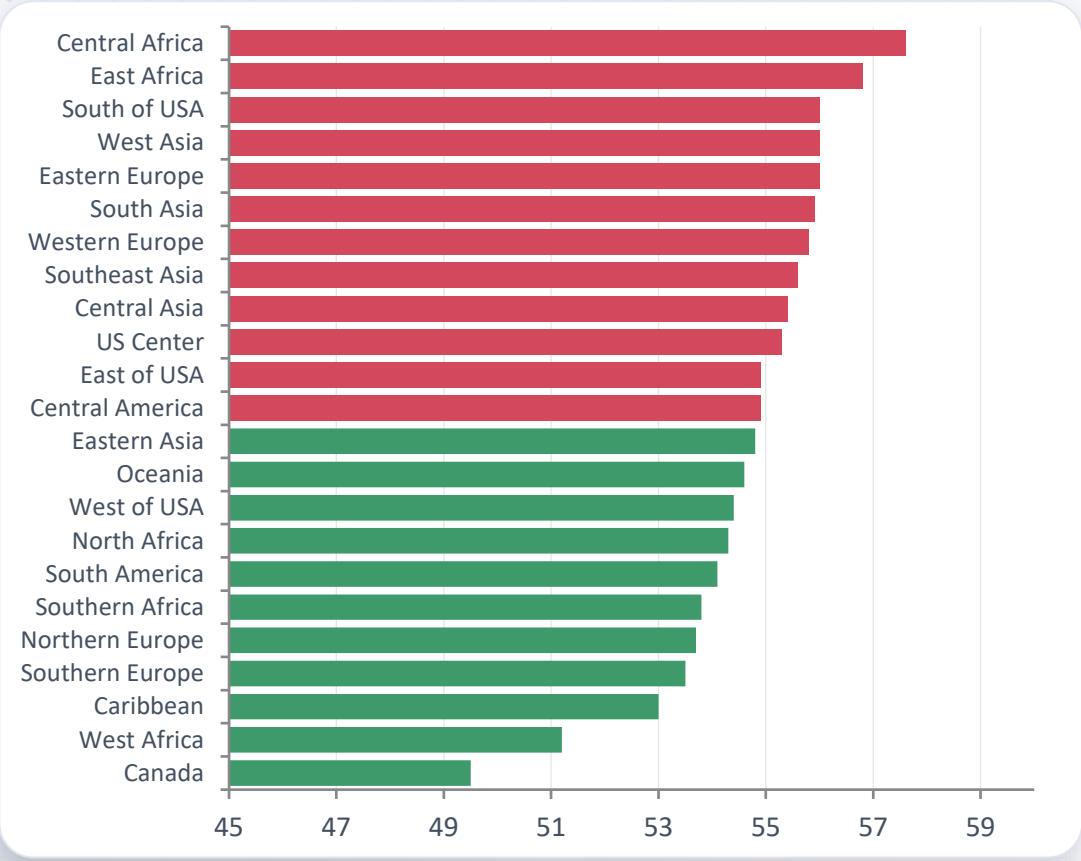
What the data shows

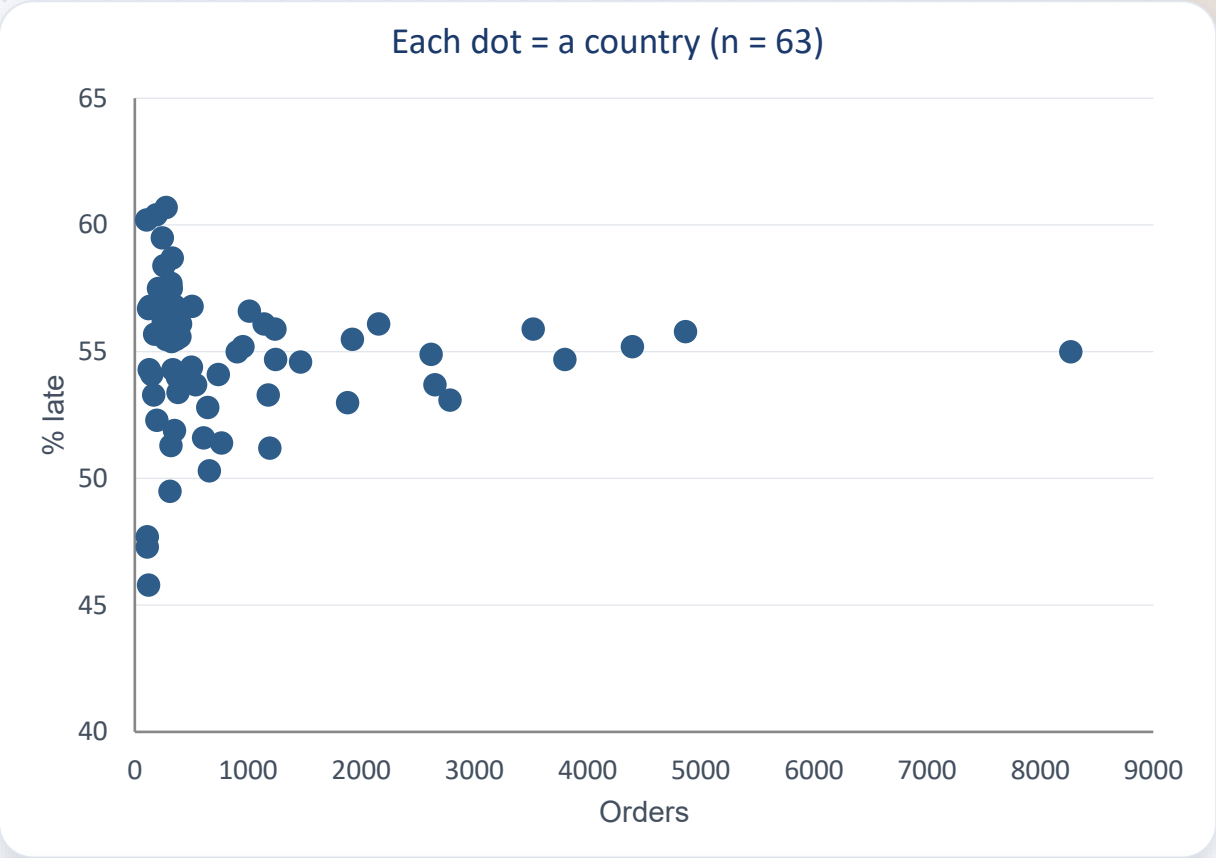
- First Class: 95.3% late
- Second Class: 76.7% late
- Standard Class: 38.1% late
- Chi-square p ≈ 0, Cramér's V = 0.456 (strong)

TAKEAWAY Fast modes promise 0–2 days but deliver in 1–4 – the promise is the problem, not just the carrier.

Objective 2: Geography barely matters

Late % by region and by country (countries with 100+ orders)

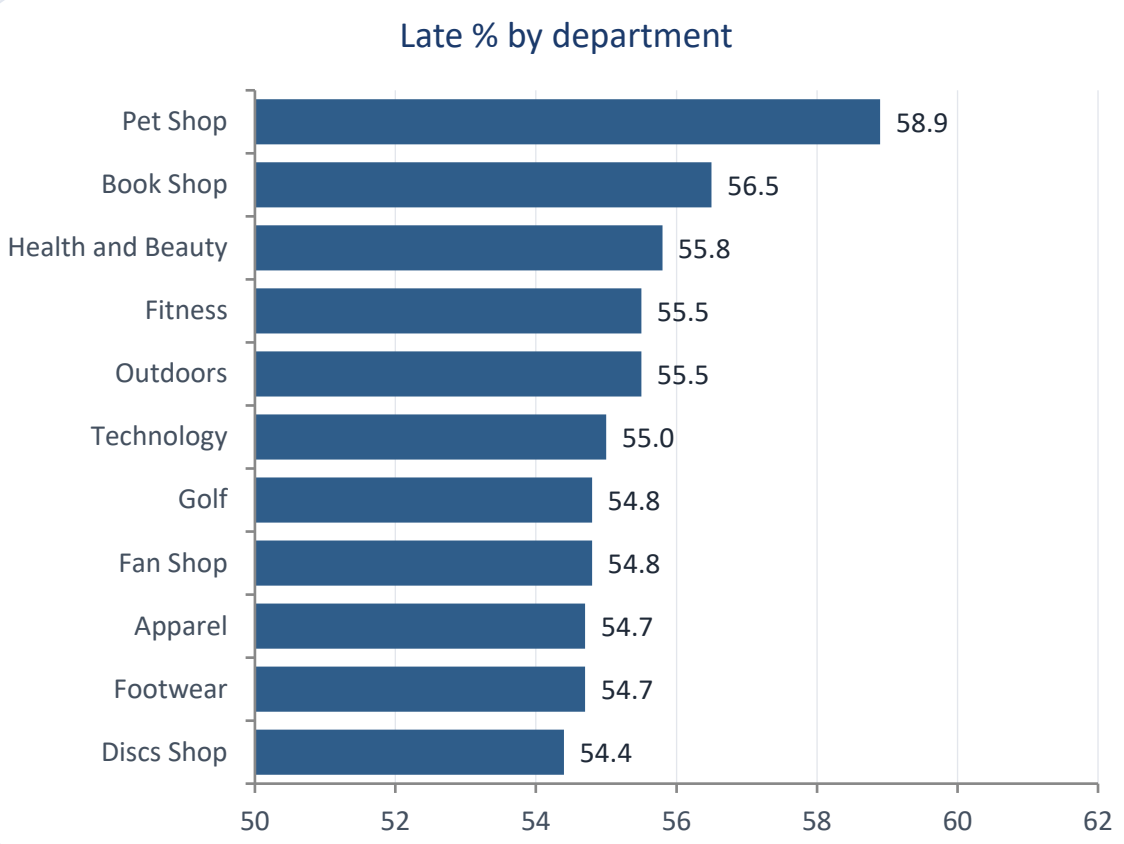


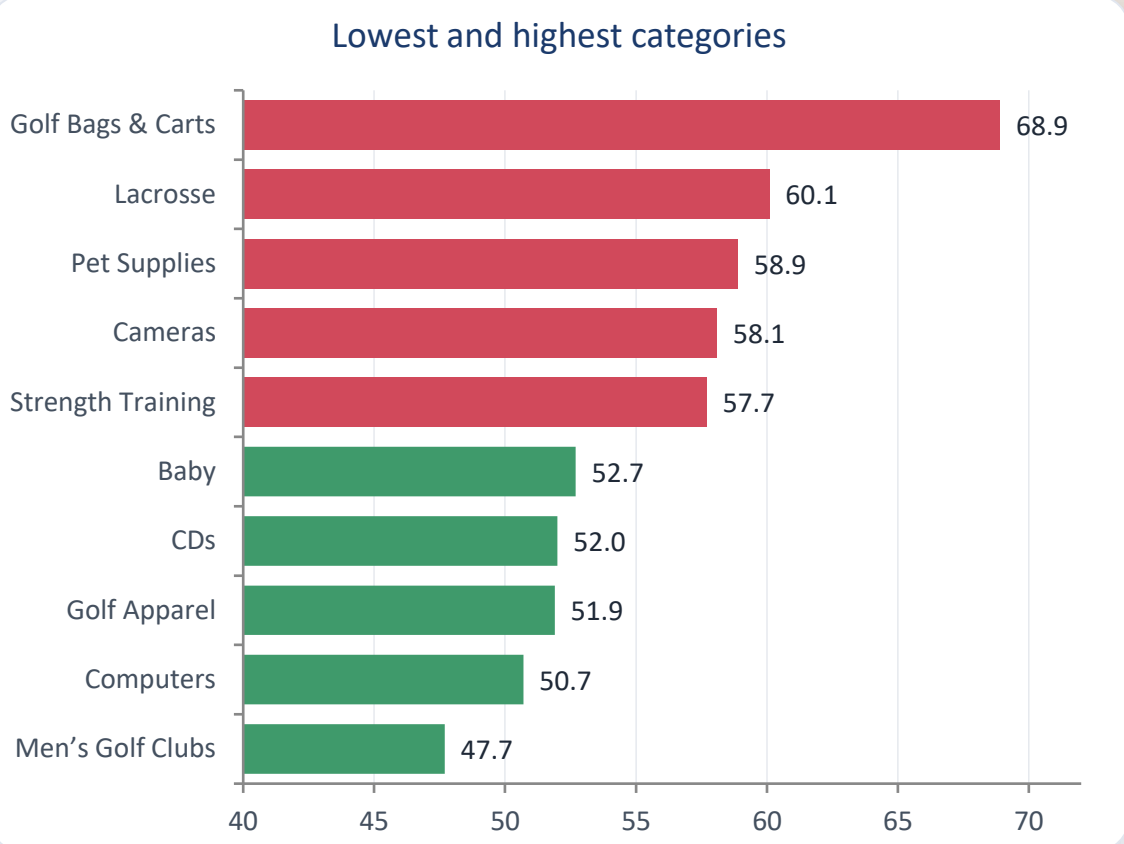


TAKEAWAY Regions span only 49.5%–57.6% (χ^2 p = 0.086, V = 0.022): not significant. Country extremes rest on small samples.

Objective 3: Product category has little effect

Item level – each item inherits its order's outcome

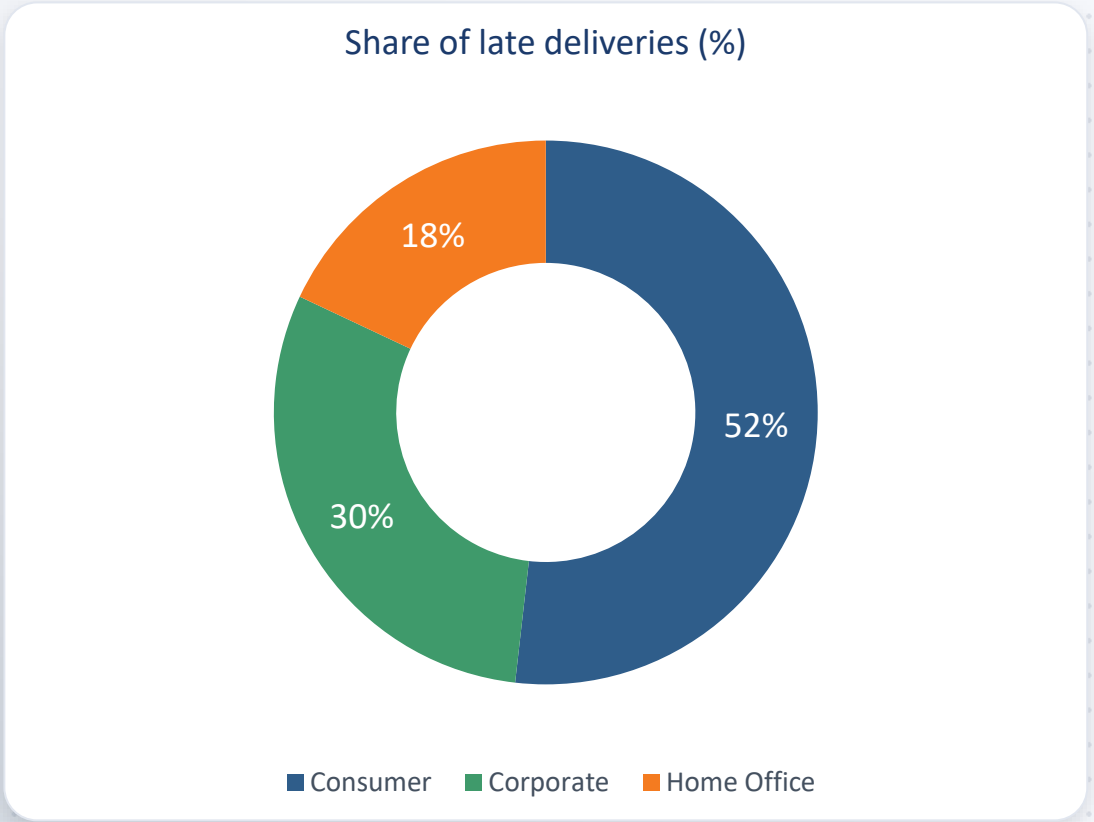


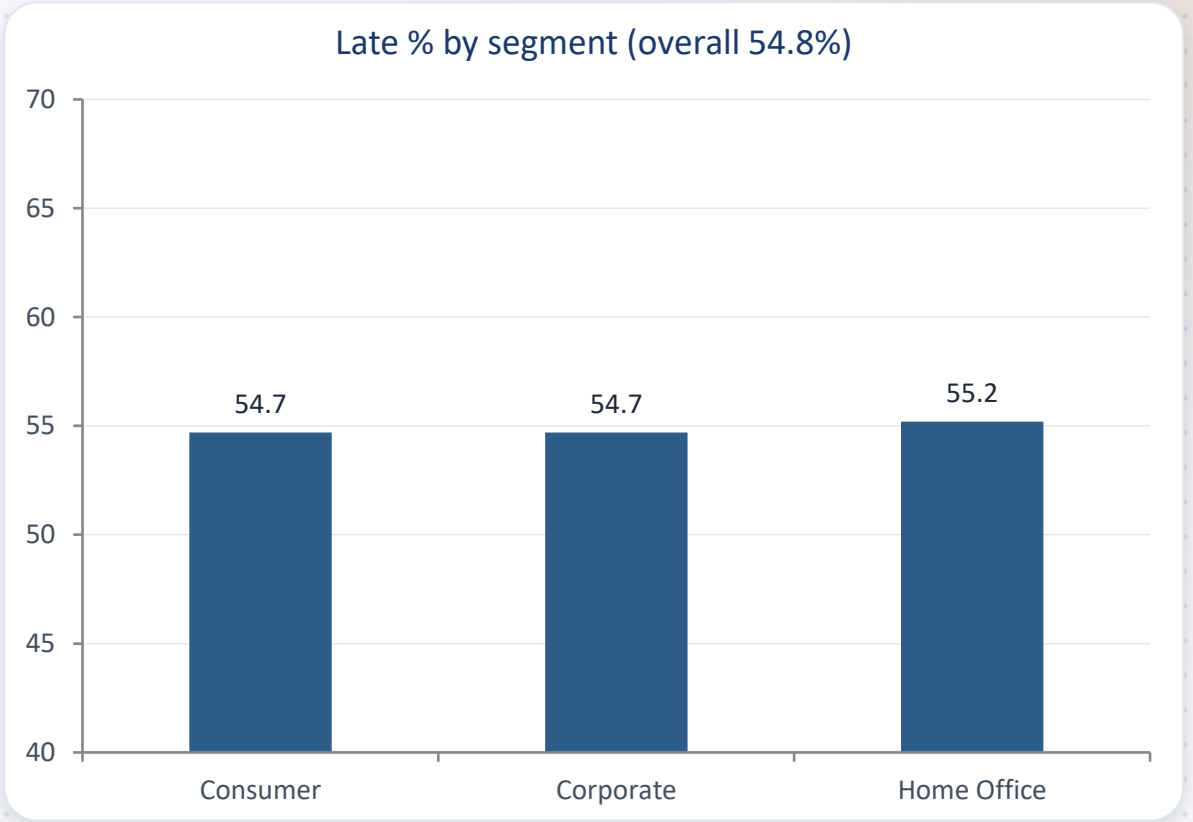


TAKEAWAY Departments span 54.4%–58.9% (χ^2 p = 0.756, V = 0.006). Golf Bags & Carts (68.9%) has only 61 items.

Objective 4: Every segment is equally exposed

Share of late orders mirrors segment size, not higher risk



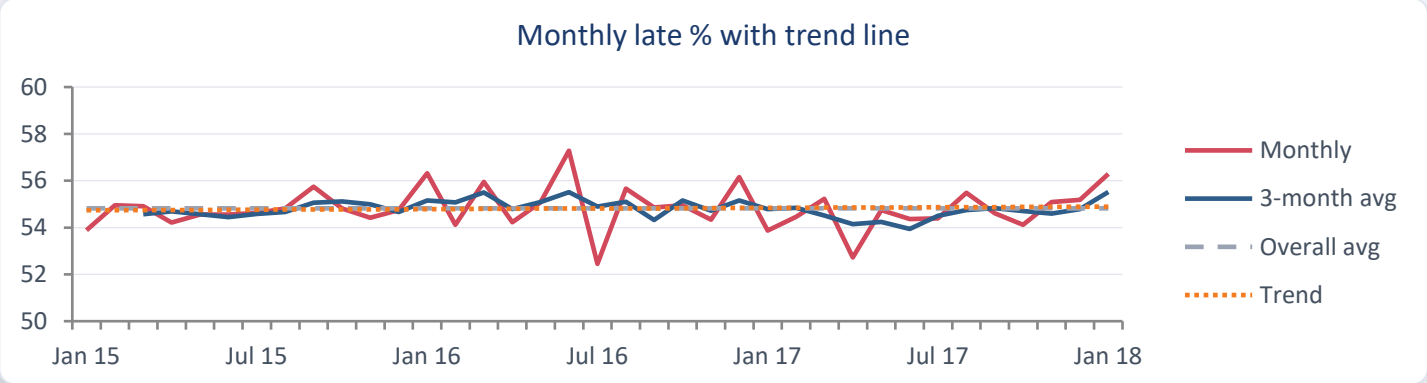


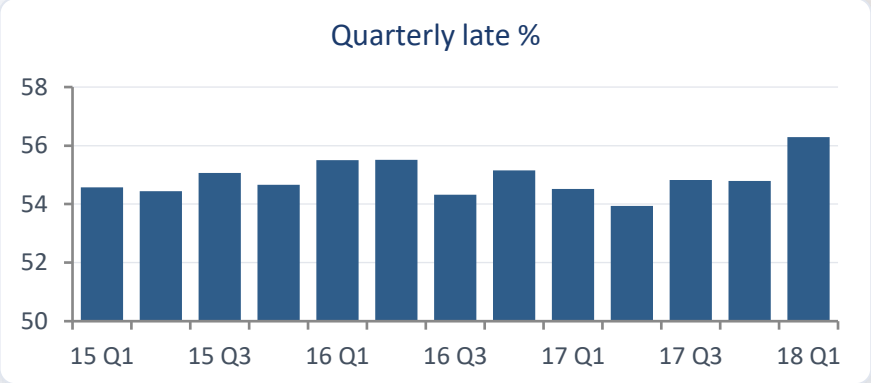
TAKEAWAY 54.7% / 54.7% / 55.2% (χ^2 p = 0.613). Customer value tier also shows no effect (p = 0.991): no case for segment-specific policy.

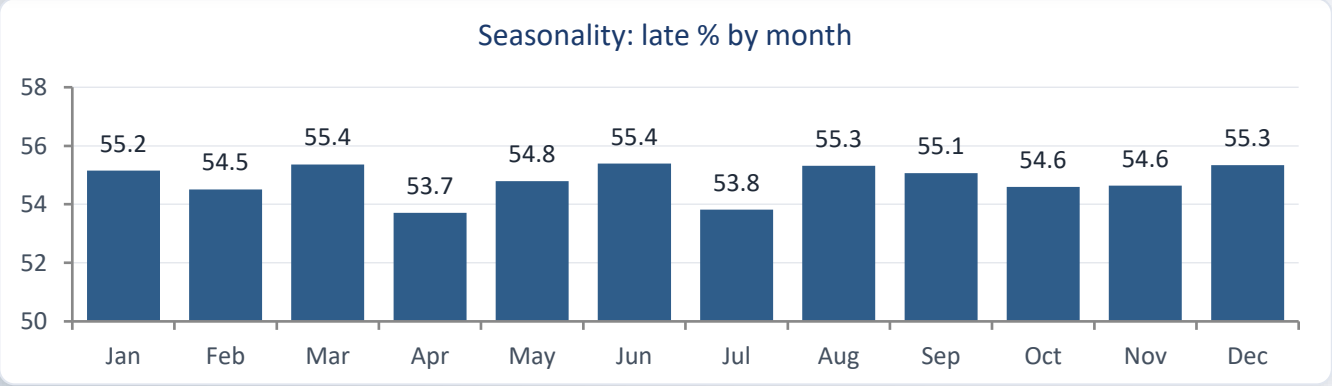
Objective 5: Temporal trends

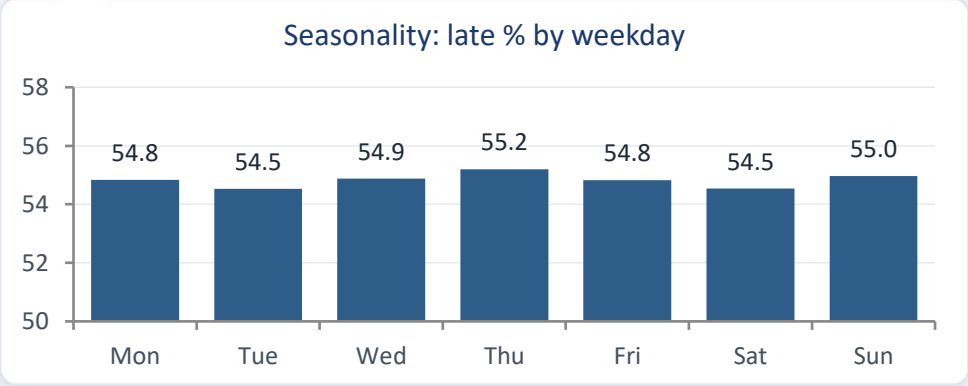
No trend and no seasonality – late % stays close to the 54.8% average

54.8%	Overall late %
56.3%	Latest month (Jan 2018)
65,752	Orders



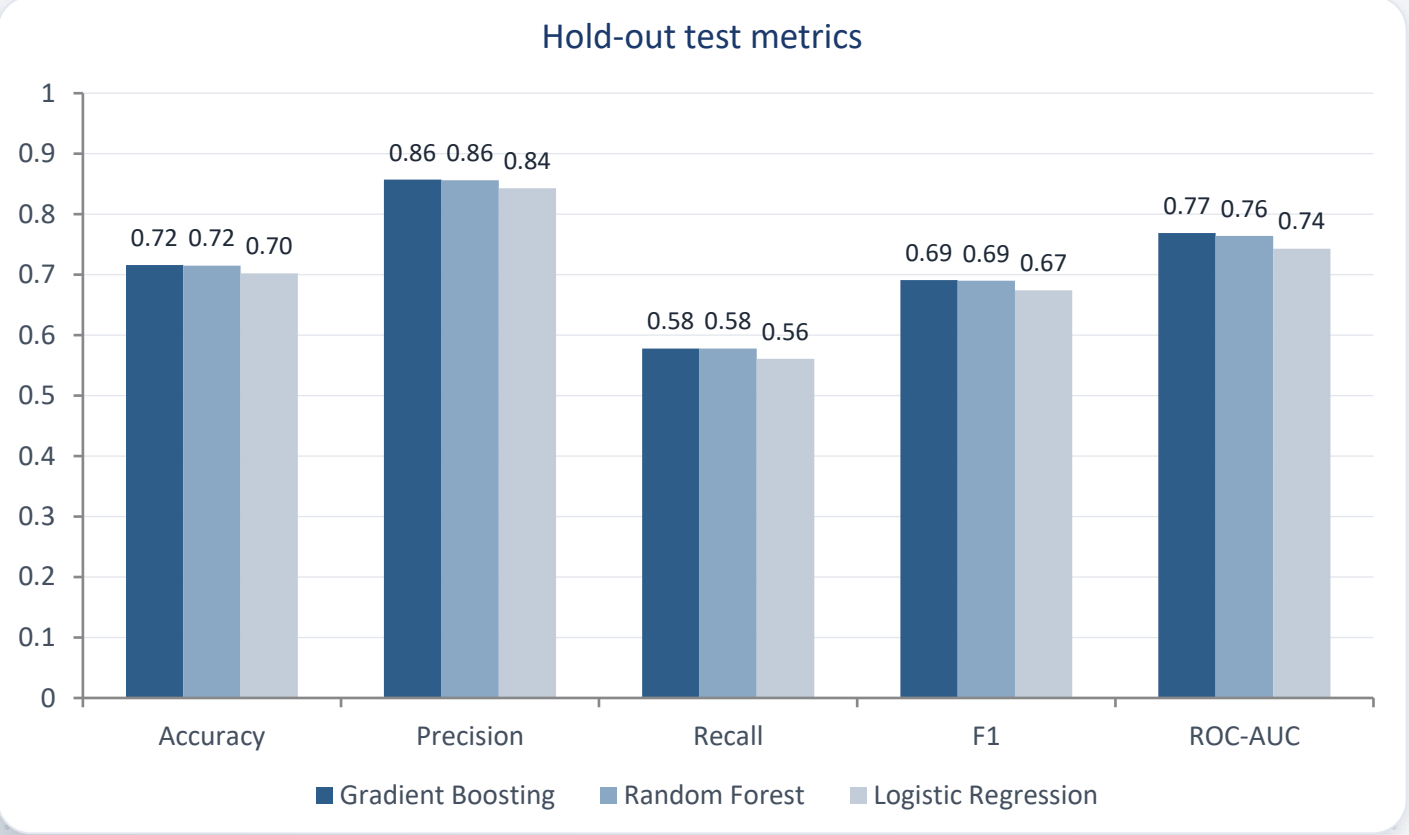






Predictive model: three models compared

80/20 stratified split; only information known at order time



Gradient Boosting – confusion matrix

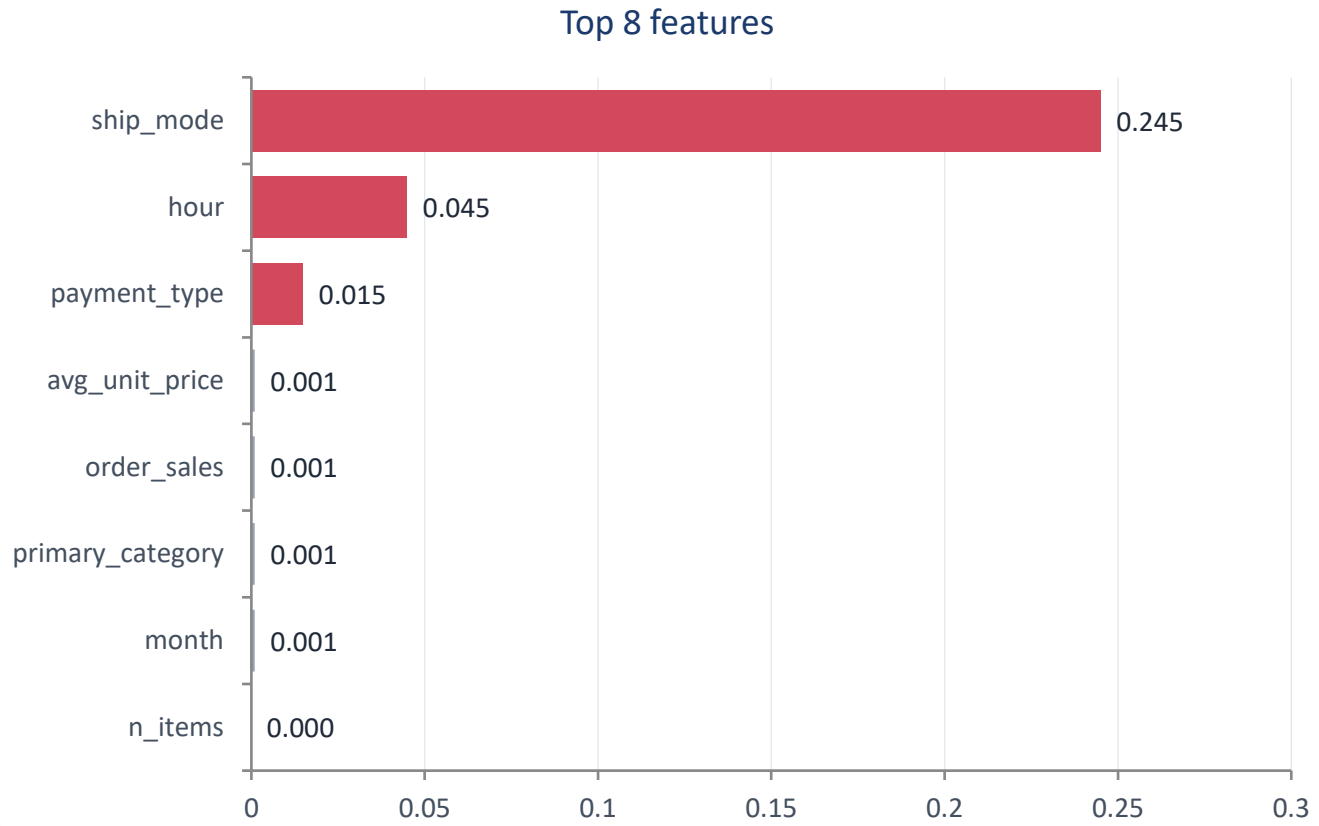
	Pred. on time	Pred. late
On time	5,243	698
Late	3,040	4,170

13,151 test orders. 5-fold CV ROC-AUC: 0.769 ± 0.004 – stable, not overfit.

TAKEAWAY **Gradient Boosting:** AUC 0.769, precision 85.7% (flags are reliable), recall 57.8% (some late orders are missed).

What drives the prediction?

Permutation importance – drop in AUC when a feature is shuffled

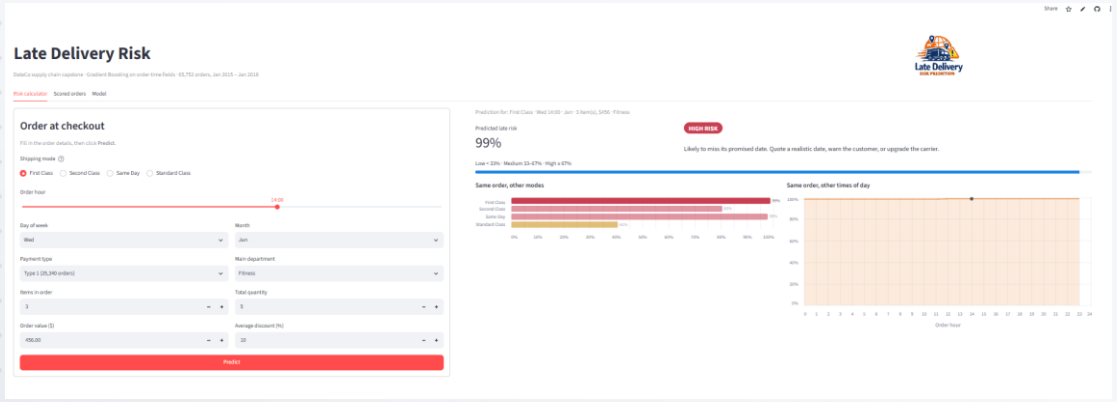


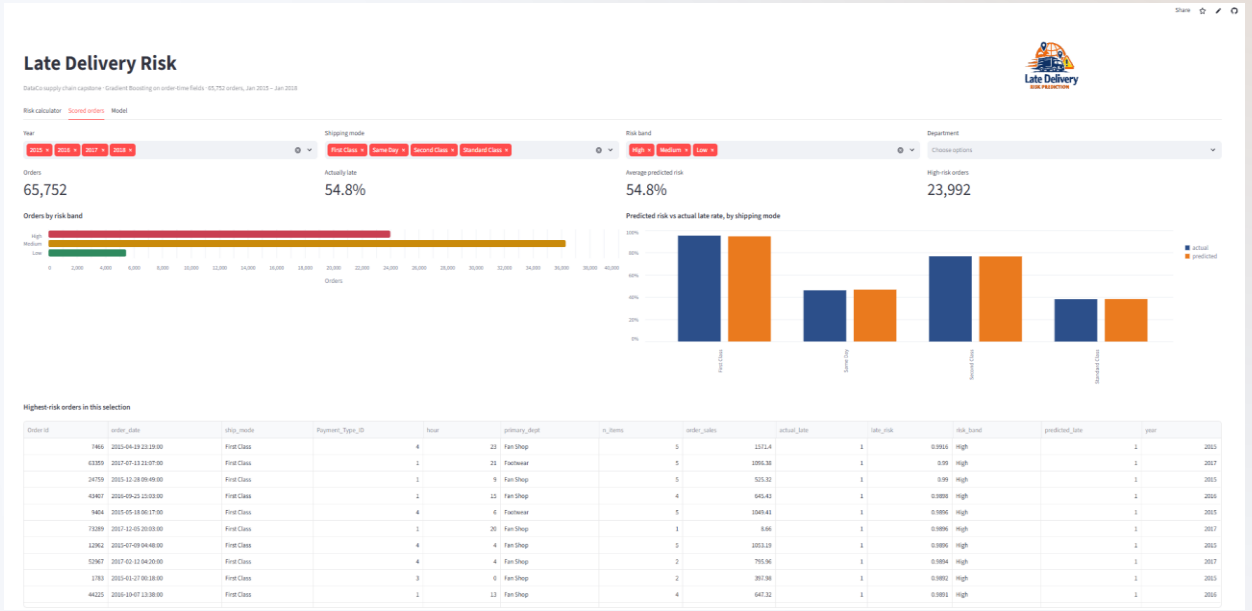
Reading it

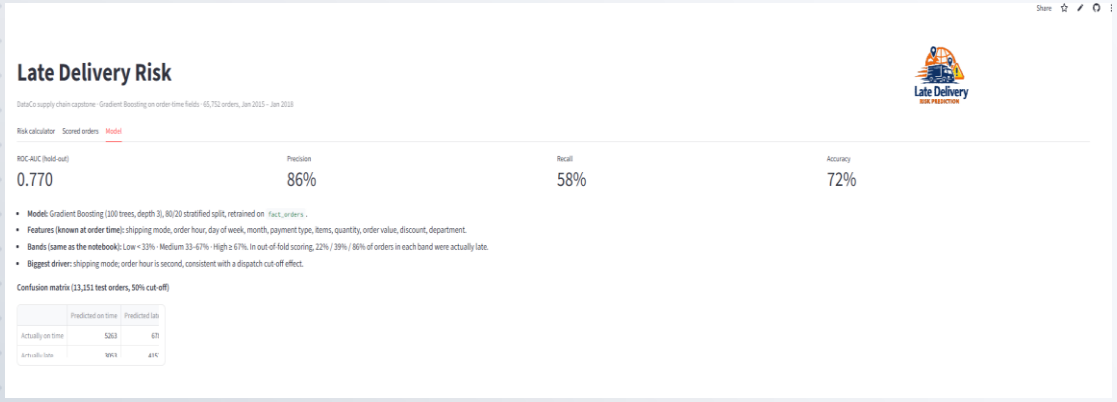
- Shipping mode (≈0.245) is by far the biggest signal
- Order hour (≈0.044): a possible cut-off or processing-time effect
- Payment type (≈0.015): minor
- Region, country, category and segment ≈ 0

TAKEAWAY The model confirms the descriptive story: how we ship matters; where and what we ship barely do.

Model Inferencing







App Link https://late-delivery-risk-prediction-dataco.streamlit.app/

Findings and recommendations

What we found, and what to do

Key findings

- **54.8% of orders are late**
  Target is balanced
- **Shipping mode is the only strong driver**
  Cramér's V = 0.456
- **Geography, category, segment, time are flat**
  No statistically meaningful effect
- **Model reaches AUC 0.77**
  Using order-time data only

Recommendations

1. **Reset delivery promises for First/Second Class**
   Promised 1–2 days vs actual 2–4 – or add carrier capacity
2. **Score every order at checkout**
   Act on high-risk orders: notify the customer or upgrade the carrier
3. **Investigate order hour and payment type**
   Cut-offs, payment holds
4. **Avoid region- or category-specific fixes**
   No evidence they pay off
5. **Enrich the data**
   Carrier, warehouse, stock, distance – to lift AUC

LIMITS One company, 2015–Jan 2018; recall of 58% means tuning the threshold to the cost of a missed late order.

Anticipated questions (1 of 2)

Suggested answers

Late Delivery
RISK PREDICTION

Questions?

15 min · keep answers to ~30 seconds

1 Why is accuracy only about 72%?

Delivery outcome depends on factors we don't have (carrier, warehouse, stock, weather). We also removed post-delivery fields to avoid leakage, so 72% accuracy and 0.77 AUC is honest, not inflated.

2 Why not use actual shipping days or delivery status?

They are only known after delivery. Including them would score near-perfectly but be useless at order time.

3 Isn't the "late" flag circular?

It matches actual > scheduled days in 97.5% of orders, so it is a genuine outcome. The model predicts it from order-time information only.

4 Recall is 58% – is that acceptable?

Precision is high (86%), so flags are reliable. To catch more late orders, lower the 0.5 threshold; the right level depends on the cost of a missed late order versus a false alarm.

5 Does geography really not matter?

Regional differences are small (49.5–57.6%, $p = 0.086$, $V = 0.022$). Extremes such as Portugal come from small samples, so we don't recommend regional fixes.

Anticipated questions (2 of 2)

Suggested answers

Late Delivery
RISK PREDICTION

Questions?

15 min · keep answers to ~30 seconds

6 **Why is First Class the worst mode?**

It promises 1 day but averages 2 – a promise the network can't meet. Standard Class promises 4 days and delivers in 4.

7 **Why Gradient Boosting?**

Highest ROC-AUC (0.769) and F1 of three models, confirmed by 5-fold CV (0.769 ± 0.004). The margin over logistic regression is modest.

8 **How would this be used operationally?**

Score each order at checkout into Low / Medium / High risk. For High risk, set realistic dates, warn the customer or upgrade the carrier.

9 **Can the results be generalised?**

Data covers one company from Jan 2015 – Jan 2018. Findings are strong for this network; revalidate on recent data before rollout.